

# KANISHKA KOPPERLA

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## SKILLS

**Machine Learning & AI:** Python, scikit-learn, TensorFlow, PyTorch, Hugging Face, OpenCV, XGBoost, JavaScript, NLP  
**LLMs & GenAI:** LangChain, LangGraph, RAG, Prompt Engineering, LLaMA, Ollama, Vector Embeddings, Semantic search  
**Backend & APIs:** FastAPI, Flask, REST APIs, Postman, Microservice Architecture, Redis  
**LLM APIs & Agentic Tools:** Claude API / Anthropic API, MCP (Model Context Protocol), AI coding tools (GitHub Copilot, Claude)  
**DevOps & Tools:** Docker, Unit & Integration Testing, Agile/Scrum, MLflow, DVC, Evidently AI, Feast, Apache Airflow, BentoML, CI/CD (GitHub Actions)  
**Database & ETL:** PostgreSQL, MongoDB, ETL Pipelines, Stored Procedures, GCP (BigQuery, Pub/Sub, Cloud Functions), Pandas, NumPy, Parquet, SQL

## EDUCATION

### Portland State University

*Masters in Computer Science*

**Sep 2024 - Present**

*GPA: 3.8*

• **Coursework:** Artificial Intelligence, Machine Learning, LLM, Data Engineering, Software Engineering

### Anna University

*Bachelor of Technology, Information Technology*

**Jun 2020 - May 2024**

*GPA: 3.5*

• **Coursework:** Data Structure and Algorithm, Agile, Database, OOPS using Python

## EMPLOYMENT HISTORY

### ATTENTION.AD

*AI Software Engineer Intern*

**Jun 2025 - Aug 2025**

*California, USA*

- Engineered AI-powered data pipelines in **Python** and **SQL** to automate structured data ingestion, transformation, and validation - supporting scalable analytics and internal reporting systems.
- Integrated machine learning models into production workflows for anomaly detection and automated dataset validation, reducing manual data review effort by **50%**.
- Designed and deployed automated data quality checks within the AI ingestion pipeline, eliminating manual validation steps across hundreds of daily records and improving pipeline reliability in Agile/Scrum sprints.

### VIZDALE

*Software Engineer*

**Jun 2022 - Feb 2024**

*Chennai, India*

- Engineered and maintained production-grade **REST APIs** using **Python** and **FastAPI**, enabling reliable data exchange across backend services and internal platforms.
- Developed LLM-powered features and **RAG** workflows integrating embeddings and vector search for contextual document intelligence, improving retrieval relevance by **35%**.
- Optimized **PostgreSQL** database queries and backend logic, reducing **API** latency by **30%** and improving overall system performance.
- Containerized backend microservices using **Docker** and maintained **CI/CD** pipelines via GitHub Actions, ensuring reliable and consistent production deployments.

## PROJECTS

### MLOps PLATFORM WITH AUTO-RETRAINING <https://github.com/Kanishka-Msd/mlops-platform>

- Built a production-grade **MLOps** platform managing the complete ML lifecycle - data versioning, experiment tracking, automated deployment, and drift monitoring using **MLflow**, **DVC**, and **Airflow**.
- Trained and evaluated **XGBoost** model achieving **ROC-AUC 0.927**, outperforming Random Forest baseline by 1.4% across 10 MLflow-tracked experiments with full feature importance analysis.
- Deployed model as a production **REST API** via BentoML and **Docker** on **Kubernetes** with blue-green and canary releases - load tested at 1,000 concurrent users with zero downtime; automated retraining via PSI-triggered **Airflow DAG** using Evidently AI.

### INTELLIGENT DOCUMENT PROCESSING (IDP) WITH RAG <https://github.com/Kanishka-Msd/idp-rag>

- Built a production-grade document intelligence system using **OCR**, **NLP**, and **LLMs** to extract structured key-value data from semi-structured documents including invoices, contracts, and forms.
- Designed embedding-based RAG pipeline with chunking strategies, pgvector for retrieval, and optimized prompt engineering - achieving 95% extraction accuracy with reduced hallucinations.
- Developed **FastAPI** microservices for document ingestion, preprocessing, and LLM-based inference; containerized with Docker and integrated **CI/CD** via GitHub Actions for production-grade deployment.

### TEXT-TO-SQL AI SYSTEM

- Built a natural language to SQL system using **LangChain** and **GPT** enabling non-technical users to query **PostgreSQL** databases via plain English - eliminating **SQL** dependency.
- Implemented a multi-step agentic orchestration loop: schema-aware prompt engineering, self-correction feedback, and automatic query retry/fix - achieving **89%** query accuracy across 25+ tables.
- Deployed as a production **REST API (FastAPI + Docker)** with query validation, error handling, and output parsing to prevent SQL injection and ensure safe database interactions.
- Integrated query validation, error handling, and output parsing pipeline to prevent SQL injection and ensure safe, reliable database interactions in production.